Datahandling & Visualization

Activity - 8

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1) Time Series & Trend :-

* No of observations (n) = 60
* Cycle length : 12 months
* No of full cycles : 5
* Overall trend : Rising
* Fitted slope Approximately 30.64 Mwh/month

2) Seasonal Pattern :-

* Peak month : month 12
* Peak Seasonal value : 656.89 Mwh
* Trough Seasonal value : -610.83 Mwh
* Trough month : month 6
* Seasional swing : 1267.72 Mwh

(3) * Consistency of Seasonality:-

* The Seasonal pattern repeats approximately every 12 months.
* Highest demand occurs in the peak month in most of 5 cycles
* ∴ The series shows a consistent Seasonal Pattern with some random variation.

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(4) * Autocorrelation :-

* Significance bound : $2/\sqrt{60} \approx 0.258$
* ACF at lag 12 : 0.759
* ACF at lag 6 : −0.870

Interpolation :- Both values exceed the significance bound in absolute value, indicating strong seasonality with a 12-month period.

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(5) * Additive Seasonality :-

* first-cycle range : 1568 MWh
* Last-cycle range : 1490 MWh
* Ratio (last/first) : 0.950
* model selected : Additive seasonality, because the seasonal swing remains approximately constant as demand increases.
* Adjusted trend slope after removing seasonality : Approximately 29.62 MWh/month.

Management advice :-

plan electricity supply around the recurring annual demand peaks & maintain sufficient capacity during high-demand months.